$$\mathcal{R}(t) = \log \frac{P(\text{ }|c)}{P(\text{ }|c+\delta c)}$$

Drift-diffusion approximation when $\delta c/c \ll 1$

$$\frac{d\mathcal{R}}{dt} = V + \sqrt{2V}\eta \leftarrow \text{Gaussian noise } \langle \eta(t)\eta(t') \rangle \sim \delta(t-t')$$

Drift = Diffusion